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Content-Centric Modelling of Virality, Reply Pressure, and User Retention on Social Media

An IVFS Compartmental ODE Model
Calibrated on the Higgs Twitter Dataset

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Abstract

We develop a deterministic compartmental ordinary differential equation model for content lifecycle dynamics on social media. The model tracks four content states—ignored, viral, fatigued, and suppressed—together with a latent reply-pressure variable and an active-user pool. We calibrate the spread block on the SNAP Higgs Twitter retweet stream, use same-dataset reply activity as a proxy for discussion pressure, and compare against an external Jigsaw toxicity reference scale. The fitted model produces a clear amplification threshold at $\alpha_{\text{crit}} \approx 0.3165$, below which viral spread is not self-sustaining. Above this threshold, stronger amplification raises long-run viral volume and discussion pressure while lowering user retention. Our main contribution is the content-centric framing: unlike most existing compartmental models for social-media dynamics, which classify users into behavioural states, we classify content items into lifecycle compartments and couple them to pressure and user attrition. The principal limitation is that the pressure block is proxy-grounded rather than directly calibrated to observed text toxicity, so the pressure parameters are only partially identified from the available data.

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1 Background

1.1 Algorithmic amplification and platform health

Social media platforms use recommendation algorithms to select and rank content for users. These algorithms are typically optimized for engagement metrics such as clicks, shares, and time-on-platform. A growing body of empirical work shows that engagement-optimized algorithms tend to amplify politically divisive, morally charged, and emotionally provocative content [**?**, **?**, **?**]. Huszár et al. [**?**] conducted a large-scale audit of Twitter's algorithmic timeline and found that the algorithm amplified content from political parties, with particularly strong effects for politically right-leaning sources in several countries. Milli et al. [**?**] showed that content promoted by engagement-based algorithms tends to be more divisive and less satisfying to users than content they would have chosen themselves. Brady et al. [**?**] demonstrated that moral-emotional language in social media posts significantly increases their diffusion, with each additional moral-emotional word raising the retweet rate by roughly 20%.

These findings point to a platform-level tradeoff. Amplification can increase short-run engagement, but it may simultaneously intensify harmful discussion dynamics, increase user fatigue, and erode long-run platform health. Beknazar-Yuzbashev et al. [**?**] provide direct experimental evidence: a field experiment showed that exposure to toxic content on social media reduced subsequent user engagement, suggesting that toxicity imposes real costs on platform participation.

1.2 Compartmental models for information spread

Compartmental models have a long history in epidemiology and have been widely applied to information dynamics. The classical SIR (Susceptible–Infected–Recovered) model partitions a population into compartments and describes transitions between them using ordinary differential equations. The analogy between disease transmission and information spread was noted early by Daley and Kendall [**?**], who proposed a rumour-spreading model based on similar compartmental structure. In the social media setting, De Domenico et al. [**?**] used the Higgs Twitter dataset to study the anatomy of a scientific rumour, fitting epidemic-style models to retweet cascades.

The DABAR modelling procedure taught in this course—*Define, Assume, Build, Assess, Report*—provides the methodological framework for our project. We define the modelling question around platform amplification tradeoffs, state our assumptions about aggregate content dynamics, build a six-state compartmental ODE system, assess it against empirical data and analytical checks, and report the results with honest attention to limitations.

The IVFS model we develop falls into the category of a *deterministic dynamical model* in the language of the course. It is a system of first-order autonomous ODEs whose state variables are continuous and whose parameters are either calibrated from data or fixed at scenario values. The model is not stochastic, not optimization-based, and not probabilistic. Its core analytical tools are equilibrium analysis, linearized stability, threshold quantities, and sensitivity indices—all of which are covered extensively in the course material.

1.3 The content-centric gap

Most existing compartmental models for social-media dynamics classify *users* into behavioural states. For example, an SVFR-type model might place users into susceptible, viewing, forwarding, and recovered compartments [**?**, **?**]. Toxicity-focused models similarly track user infection states, where "infection" represents adoption of toxic behaviour [**?**, **?**]. These models are valuable, but

they require assumptions about individual user decision-making and local network structure that are difficult to validate at platform scale.

Our model takes a different approach. Instead of compartmentalizing users, we compartmentalize *content items* into lifecycle states: content is created, may go viral, eventually fatigues, and may be moderated. This content-centric framing has several advantages. First, it operates at the aggregate level where platform data is most readily available—retweet counts, reply volumes, and moderation actions are all content-level observables. Second, it separates the amplification mechanism (which acts on content visibility) from the user response (which is captured by a single aggregate user pool). Third, it allows us to study the platform health tradeoff directly in terms of content lifecycle dynamics rather than individual user psychology.

The price of this simplification is that we lose resolution on user heterogeneity and network topology. We treat users as a single aggregate pool U with constant replenishment and pressure-dependent attrition. This is a deliberate modelling choice: it keeps the system tractable (six states rather than dozens) while still capturing the feedback loop between content amplification, discussion pressure, and platform health.

2 Previous mathematical models

We briefly review the main lines of mathematical modelling relevant to our work. Our purpose is not to provide an exhaustive survey, but to position our IVFS model relative to the existing literature and to identify the specific gap it addresses.

2.1 Rumour models and epidemic analogies

The idea that information spreads like a disease dates to Daley and Kendall [**?**], who proposed a stochastic model for rumour transmission in which "spreaders" contact "ignorants" and either transmit the rumour or become "stiflers." This model has a natural compartmental structure analogous to SIR, and its threshold behaviour—rumours persist only when the contact rate exceeds a critical value—mirrors the basic reproduction number $\mathcal{R}_0$ in epidemiology.

De Domenico et al. [**?**] applied epidemic-style modelling to the same Higgs Twitter dataset that we use. They studied the temporal evolution of the retweet cascade around the Higgs boson discovery announcement and fitted several spreading models to the data. Their work demonstrates that compartmental models can capture the burst-and-decay dynamics of real social media cascades, which is one of the empirical motivations for our project.

2.2 User-centric compartmental models

Several recent papers have developed compartmental ODE models specifically for social media dynamics, but with *users* as the compartmentalized entities.

Li and Shao [**?**] proposed an SVFR (Susceptible–Viewing–Forwarding–Recovered) model for viral video sharing. In their framework, users transition from viewing to forwarding based on content attractiveness, and advertising interventions can modify transition rates. The model is fitted to video-sharing data and produces threshold conditions for viral spread. Jing et al. [**?**] developed a similar SIR-based model for advertising spread on social networks, incorporating network structure through degree distributions.

These user-centric models provide useful threshold and stability results, but they require assumptions about individual user states that are difficult to observe directly at platform scale. For

example, classifying a user as "susceptible" or "recovered" requires knowledge of their exposure history, which is typically not available in public datasets.

2.3 Toxicity propagation models

A separate line of work models the propagation of toxic or harmful content on social media using epidemic-style compartmental structures.

Maleki et al. [**?**] applied an epidemiological model to evaluate toxicity propagation on Twitter. Their model classifies users into compartments based on their toxicity exposure and production, and they calibrate it using text-based toxicity scores from Twitter data. Addai et al. [**?**] extended this approach by modelling interventions (content moderation, counter-speech, and user education) as control variables in a toxicity-propagation ODE system and studying their effects through simulation.

Both of these models use direct text-based toxicity labels, which is an advantage over our proxy-based approach. However, both also compartmentalize *users* rather than content, and neither explicitly models the feedback loop between content amplification, discussion pressure, and user retention that is central to our model.

2.4 Positioning of the present model

Table 1 summarizes the key distinctions. Our IVFS model differs from the prior work in three respects: (i) it compartmentalizes content lifecycle states rather than user behavioural states, (ii) it explicitly couples content dynamics to a latent discussion-pressure variable and an active-user pool, and (iii) it is calibrated on a real event cascade (the Higgs dataset) rather than on simulated or aggregate data alone.

Table 1: Comparison of compartmental modelling approaches for social media dynamics.

Model	Compartments	Pressure	User feedback	Calibration
Daley–Kendall [**?**]	Users	No	No	Analytical
Li–Shao SVFR [**?**]	Users	No	No	Video sharing data
Jing et al. SIR [**?**]	Users	No	No	Simulated
Maleki et al. [**?**]	Users	Toxicity labels	No	Twitter toxicity
Addai et al. [**?**]	Users	Toxicity labels	No	Simulated
This paper (IVFS)	**Content**	**Reply proxy**	**Yes**	**Higgs retweet data**

The main innovation is the content-centric perspective. By tracking content items through lifecycle stages rather than users through behavioural stages, we obtain a model whose state variables correspond directly to platform-level observables (content volumes, reply rates, active user counts) rather than to latent user states that must be inferred.

3 Modelling question

The central question we address is:

How does platform amplification affect the lifecycle of content, the intensity of harmful discussion, and the long-run size of the active user base?

We also investigate three subsidiary questions:

1. At what amplification level does viral spread become self-sustaining, and what is the nature of this threshold?
2. How sensitive are the long-run outcomes to the parameters governing the pressure block?
3. Can one dataset (the SNAP Higgs Twitter activity stream) support both a spread calibration and a discussion-pressure proxy without overclaiming direct toxicity measurement?

These questions are well-suited to a compartmental ODE framework because they involve threshold behaviour, equilibrium comparisons, and parameter sensitivity—all standard tools in the dynamical-systems portion of the course.

4 Modelling approach

We use a deterministic six-state compartmental ordinary differential equation (ODE) system. This is a *deterministic dynamical model* in the classification used in the course: the state evolves continuously in time according to a system of first-order ODEs, and there is no stochastic or probabilistic component.

The modelling approach follows the DABAR procedure:

1. **Define.** We define the modelling question around the amplification–health tradeoff, with content lifecycle states as the primary dynamical variables.
2. **Assume.** We assume that aggregate content and user activity can be approximated by continuous deterministic flows, that content items can be grouped into four lifecycle compartments, and that harmful discussion pressure can be summarized by a single latent scalar state.
3. **Build.** We construct a six-state ODE system with content compartments (I, V, F, S), a pressure state τ, and a user pool U, with couplings that capture the amplification feedback loop.
4. **Assess.** We calibrate the spread block against the Higgs Twitter dataset, benchmark against a classical SIR alternative, validate on multiple time windows, and perform sensitivity and robustness analyses.
5. **Report.** We present the results with clear attention to what is well-identified (spread parameters) and what is only partially identified (pressure parameters).

The empirical strategy has two stages. First, we calibrate the spread block using a reduced IVF helper model fitted to the Higgs retweet stream. This stage identifies the spread-side parameters $(\beta_0, \gamma_0, \lambda_0, d_\lambda)$ with narrow confidence intervals. Second, we simulate the full six-state IVFS model under different amplification scenarios to study the policy tradeoff. The pressure block is handled separately: same-dataset reply activity (RE) serves as a proxy for latent discussion pressure, and the Jigsaw toxicity dataset provides an external reference scale for the magnitude of τ, but neither source uniquely pins down the pressure parameters (ϕ, ψ).

This two-stage approach is a deliberate modelling choice. It means the spread calibration is clean and well-identified, while the pressure block is honestly reported as partially constrained. We believe this is preferable to forcing a joint calibration that would conflate well-identified and poorly-identified parameters.

5 Formulation of the model

5.1 State variables

The IVFS model is a deterministic compartmental ODE system with six state variables. The first four represent content lifecycle compartments, the fifth captures latent discussion pressure, and the sixth tracks the aggregate active-user pool.

Variable	Interpretation
$I(t)$	Ignored or not-yet-viral content. This compartment represents content items that have been posted on the platform but have not yet achieved significant visibility. Content enters the system through I.
$V(t)$	Viral content. Content items that have achieved high visibility and are being actively shared and engaged with. This is the compartment most directly affected by the amplification parameter α.
$F(t)$	Fatigued content. Content that was previously viral but has lost engagement momentum. Users have moved on, and the content is no longer being actively shared.
$S(t)$	Suppressed or moderated content. Content that has been removed or de-ranked by platform moderation. In the current model, moderation acts only on ignored content ($I \to S$).
$\tau(t)$	Latent discussion pressure. A scalar state variable representing the aggregate intensity of harmful, heated, or polarized discussion on the platform. This is not directly observed text toxicity; it is a latent variable driven by viral content and relaxing over time.
$U(t)$	Active users. The aggregate number of users actively participating on the platform. This pool is replenished at a constant rate and depleted at a rate that increases with discussion pressure.

The four content compartments (I, V, F, S) form a lifecycle: content is created (enters I), may go viral (moves to V), eventually fatigues (moves to F), and may be moderated at any point (moves to S). All content also decays naturally at rate μ_c, representing loss of visibility over time. This lifecycle structure is the core of the model and is what distinguishes it from user-centric approaches.

5.2 Parameters

Table 2 lists all parameters in the model. We distinguish between *calibrated parameters*, which are fitted to data, and *scenario parameters*, which are fixed at values chosen to represent different policy configurations.

Table 2: IVFS model parameters.

Parameter	Value	Type	Interpretation
α	varies	Policy	Amplification level (platform tuning knob)
β_0	1.000073	Calibrated	Base spread rate (fitted from Higgs RT)
γ_0	0.300283	Calibrated	Base fatigue rate (fitted from Higgs RT)
λ_0	0.199996	Calibrated	Exogenous seed amplitude
d_λ	0.044641	Calibrated	Exogenous seed decay rate
κ	0.8	Scenario	Pressure effect on amplification
η	0.3	Scenario	Pressure effect on fatigue
ϕ	0.056	Scenario	Viral-to-pressure forcing rate
ψ	0.1	Scenario	Pressure relaxation rate
ρ	0.06	Scenario	Content injection rate per user
λ_u	0.02	Scenario	User attrition rate
ν	1.0	Scenario	User replenishment rate
μ_c	0.01	Scenario	Natural content decay rate
δ	0.05	Scenario	Moderation / suppression rate
w	10.0	Scenario	Pressure effect on user attrition

The amplification parameter $\alpha \in [0, 1]$ is the key policy variable. It represents the degree to which the platform's recommendation algorithm promotes engaging content. Setting α close to 1 corresponds to aggressive engagement optimization; setting α close to 0 corresponds to a health-first policy that limits algorithmic amplification.

5.3 Modelling assumptions

The formulation rests on the following assumptions. We state each assumption explicitly and discuss its justification.

1. **Continuous deterministic flows.** Aggregate content and user activity can be approximated by continuous variables evolving according to deterministic ODEs. This is a standard assumption in compartmental modelling and is justified when the population sizes are large enough that stochastic fluctuations are negligible. For the Higgs dataset, the total event volume (563,069 events) supports this approximation.

2. **Content lifecycle compartments.** Content items can be meaningfully grouped into four lifecycle stages: ignored, viral, fatigued, and suppressed. The boundaries between these stages are not sharp in reality, but the compartmental approximation captures the essential dynamics of content creation, amplification, decay, and removal.

3. **Homogeneous mixing.** We assume that all content items in a given compartment have equal probability of transitioning to other compartments. This is the standard well-mixed assumption in compartmental models. It ignores network structure, community effects, and content heterogeneity. We discuss the consequences of this assumption in Section **??**.

4. **Scalar pressure.** Harmful discussion dynamics are summarized by a single scalar state τ. In reality, online toxicity is multidimensional (hate speech, harassment, misinformation, etc.), but a single-index representation keeps the model tractable and still captures the essential feedback loop.

5. **Aggregate user pool.** The active-user population is a single aggregate pool with constant replenishment and pressure-dependent attrition. We do not model user heterogeneity, user types, or network-dependent behaviour.

6. **Local parameter constancy.** The spread-side parameters (β_0, γ_0) are treated as locally constant on the 100-hour calibration window. The additional 48-hour window validation (Section **??**) shows that the model shape is robust across windows, even though the parameter values shift.

5.4 The IVFS equations

The model is the following system of six coupled first-order ODEs:

$$\dot{I} = \underbrace{\frac{\rho U}{1+U}}_{\text{content creation}} - \underbrace{\beta_{\text{eff}}\, I\, V}_{\text{viral conversion}} - \underbrace{(\delta + \mu_c)\, I}_{\text{moderation + decay}}, \tag{1}$$

$$\dot{V} = \underbrace{\beta_{\text{eff}}\, I\, V}_{\text{viral conversion}} - \underbrace{\gamma_{\text{eff}}\, V}_{\text{fatigue}} - \underbrace{\mu_c\, V}_{\text{decay}}, \tag{2}$$

$$\dot{F} = \underbrace{\gamma_{\text{eff}}\, V}_{\text{fatigue inflow}} - \underbrace{\mu_c\, F}_{\text{decay}}, \tag{3}$$

$$\dot{S} = \underbrace{\delta\, I}_{\text{moderation}} - \underbrace{\mu_c\, S}_{\text{decay}}, \tag{4}$$

$$\dot{\tau} = \underbrace{\phi\, V}_{\text{pressure forcing}} - \underbrace{\psi\, \tau}_{\text{pressure relaxation}}, \tag{5}$$

$$\dot{U} = \underbrace{\nu}_{\text{replenishment}} - \underbrace{\lambda_u (1 + w\tau)\, U}_{\text{pressure-dependent attrition}}, \tag{6}$$

where the effective rates are

$$\beta_{\text{eff}} = \alpha\, \beta_0\, (1 + \kappa\, \tau), \qquad \gamma_{\text{eff}} = \gamma_0\, (1 + \eta\, \tau). \tag{7}$$

We now explain each equation in turn.

Content creation (Eq. 1). New content enters the ignored compartment at rate $\rho U/(1+U)$. This is a saturating (Michaelis–Menten type) inflow: when the user pool U is small, content creation scales roughly linearly with U; when U is large, creation saturates toward the maximum rate ρ. A purely linear inflow ρU made the content input unrealistically large in our simulations, which is why we adopted the saturating form. Content leaves the ignored compartment either by going viral (at rate $\beta_{\text{eff}} IV$, which is proportional to the product of ignored and viral content, analogous to the mass-action transmission term in SIR models) or by being moderated (δI) or decaying naturally ($\mu_c I$).

Viral dynamics (Eq. 2). Content enters the viral compartment through the mass-action conversion term $\beta_{\text{eff}} IV$. The effective spread rate $\beta_{\text{eff}} = \alpha\beta_0(1+\kappa\tau)$ captures two effects: the platform amplification parameter α scales the base rate β_0, and the pressure feedback $(1+\kappa\tau)$ means that

higher discussion pressure increases the rate at which content goes viral. This is the core of the amplification feedback loop. Viral content leaves V by fatiguing at rate $\gamma_{\text{eff}}V$ or decaying at rate $\mu_c V$.

Fatigue and suppression (Eqs. 3–4). These are accumulation compartments. Fatigued content collects previously viral items that have lost momentum; suppressed content collects items removed by moderation. Neither F nor S feeds back into the spread-pressure-user dynamics. They are "sinks" in the feedback structure, but they are important for interpreting the content lifecycle and for total content accounting.

Discussion pressure (Eq. 5). The latent pressure state τ is driven by viral content at rate ϕV and relaxes at rate $\psi\tau$. This is a simple first-order linear response: pressure builds when viral content is present and decays exponentially when it is not. The equilibrium pressure level (when V is constant) is $\tau^* = \phi V^*/\psi$, so the ratio ϕ/ψ sets the scale of long-run pressure.

User dynamics (Eq. 6). Users join the platform at a constant rate ν and leave at rate $\lambda_u(1 + w\tau)U$. The key feature is the pressure-dependent attrition: higher discussion pressure causes faster user departure. At equilibrium, $U^* = \nu/[\lambda_u(1 + w\tau^*)]$, so increased pressure directly reduces the steady-state user base.

Effective rates (Eq. 7). The effective spread rate β_{eff} and effective fatigue rate γ_{eff} both depend on the pressure τ. Higher pressure increases both rates: it makes content spread faster (viral amplification) but also fatigue faster (burnout effect). This dual dependence creates a nonlinear feedback: amplification drives viral content, which drives pressure, which in turn modifies both the amplification and fatigue rates.

5.5 Structure of the feedback loop

Figure 1 shows the state-flow structure of the model. The essential feedback loop is:

$$V \xrightarrow{\phi} \tau \xrightarrow{\kappa} \beta_{\text{eff}} \to V.$$

Viral content drives pressure, pressure amplifies the effective spread rate, and the amplified spread rate produces more viral content. This positive feedback loop is counteracted by fatigue ($V \to F$), pressure relaxation ($\psi\tau$), and pressure-dependent user attrition ($\lambda_u w\tau U$), which reduces the content creation rate.

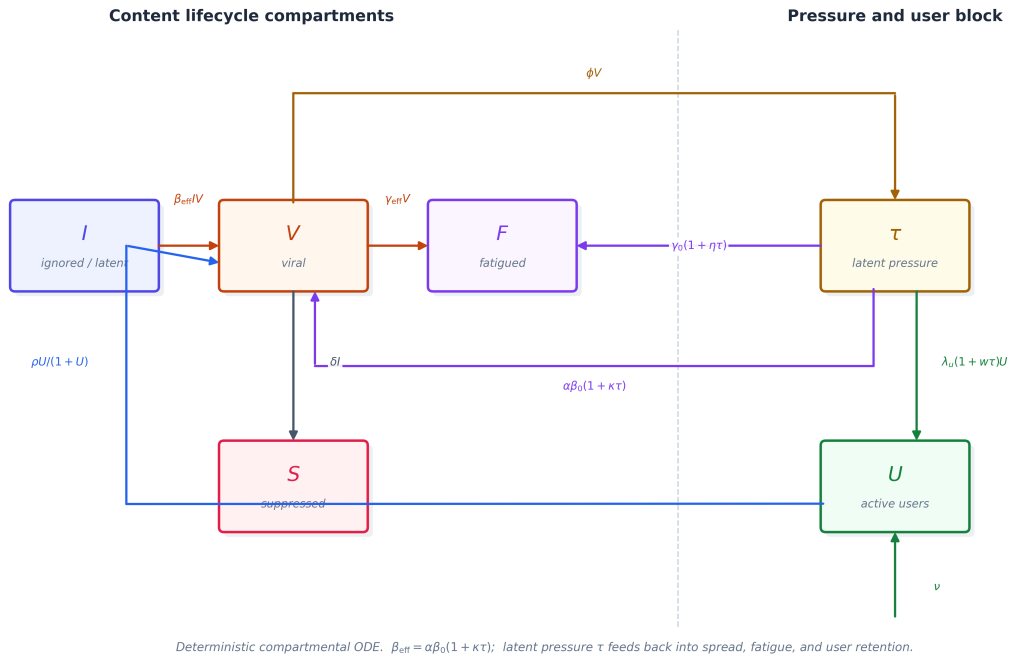

Figure 1: State-flow structure of the IVFS model. Content items move through ignored, viral, fatigued, and suppressed compartments. Latent discussion pressure feeds back into spread and fatigue rates and drives user attrition.

5.6 Threshold quantity and disease-free equilibrium

A central quantity in any compartmental model is the threshold that determines whether the "infection" (in our case, viral content) can persist in the long run. Following standard practice in epidemic modelling, we compute this threshold by linearizing around the disease-free equilibrium (DFE).

At the DFE, there is no viral content: $V = 0$, $\tau = 0$, $F = 0$. The remaining equations give:

$$U_{\text{DFE}} = \frac{\nu}{\lambda_u}, \qquad I_{\text{DFE}} = \frac{\rho\, U_{\text{DFE}}}{(1 + U_{\text{DFE}})(\delta + \mu_c)}. \tag{8}$$

With the current parameter values, $U_{\text{DFE}} = 50.0$ and $I_{\text{DFE}} \approx 0.9804$.

Linearizing the $\dot{V}$ equation around the DFE yields

$$\dot{V} \approx \big(\alpha\, \beta_0\, I_{\text{DFE}} - \gamma_0 - \mu_c\big) V.$$

Viral content grows when the coefficient is positive and decays when it is negative. We define the basic reproduction number:

$$\boxed{\mathcal{R}_0 = \frac{\alpha\, \beta_0\, I_{\text{DFE}}}{\gamma_0 + \mu_c}.} \tag{9}$$

Viral content is self-sustaining when $\mathcal{R}_0 > 1$ and dies out when $\mathcal{R}_0 < 1$. Solving $\mathcal{R}_0 = 1$ for α gives the critical amplification level:

$$\alpha_{\text{crit}} = \frac{\gamma_0 + \mu_c}{\beta_0\, I_{\text{DFE}}} \approx 0.3165. \tag{10}$$

This threshold divides the amplification parameter space into two regimes:

- $\alpha < \alpha_{\text{crit}}$: the DFE is stable, viral content dies out, and $\tau \to 0$.
- $\alpha > \alpha_{\text{crit}}$: the DFE is unstable, and a positive (endemic) equilibrium with $V^* > 0$ and $\tau^* > 0$ exists.

We emphasize that this $\mathcal{R}_0$ is computed from linearization of the $\dot{V}$ equation at the DFE. A full next-generation matrix derivation would strengthen this result; we discuss this as a future extension in Section **??**.

6 Analysis of the model

This section presents the analytical results we have obtained for the IVFS system. We establish nonnegativity, boundedness, the DFE threshold, local stability, and sensitivity indices.

6.1 Nonnegativity

Proposition 1 (Nonnegativity)**.** *If the initial state satisfies $I(0), V(0), F(0), S(0), \tau(0), U(0) \geq 0$, then the solution remains componentwise nonnegative for as long as it exists.*

Proof. We verify that each coordinate hyperplane is inward-pointing or tangent.

- At $I = 0$: $\dot{I} = \rho U/(1+U) \geq 0$, since $U \geq 0$.
- At $V = 0$: $\dot{V} = 0$, since both the conversion and fatigue terms vanish.
- At $F = 0$: $\dot{F} = \gamma_{\text{eff}} V \geq 0$, since $V \geq 0$ and $\gamma_{\text{eff}} > 0$.
- At $S = 0$: $\dot{S} = \delta I \geq 0$, since $I \geq 0$ and $\delta > 0$.
- At $\tau = 0$: $\dot{\tau} = \phi V \geq 0$, since $V \geq 0$ and $\phi > 0$.
- At $U = 0$: $\dot{U} = \nu > 0$, since $\nu > 0$.

Since the vector field points inward (or is tangent) on every face of the nonnegative orthant $\mathbb{R}^6_{\geq 0}$, the orthant is positively invariant. □

6.2 Boundedness

Proposition 2 (Bounded invariant region)**.** *Solutions of the IVFS system with nonnegative initial data enter a bounded positively invariant region in finite time.*

Proof. We bound each group of variables in turn.

User pool. From (6), $\dot{U} = \nu - \lambda_u(1 + w\tau)U \leq \nu - \lambda_u U$, since $\tau \geq 0$. By the comparison lemma, $U(t) \leq \max\{U(0), \nu/\lambda_u\}$ for all $t \geq 0$. Denote $U_{\max} = \nu/\lambda_u$.

Total content mass. Define $C = I + V + F + S$. Summing equations (1)–(4), the internal transfer terms ($\beta_{\text{eff}} IV$, $\gamma_{\text{eff}} V$, δI) cancel, leaving:

$$\dot{C} = \frac{\rho U}{1+U} - \mu_c C \leq \rho - \mu_c C.$$

Hence $C(t) \leq \max\{C(0), \rho/\mu_c\}$. Denote $C_{\max} = \rho/\mu_c$.

Pressure. Since $V \leq C \leq C_{\max}$, equation (5) gives $\dot{\tau} \leq \phi C_{\max} - \psi\tau$, so $\tau(t) \leq \max\{\tau(0), \phi C_{\max}/\psi\}$.

All six state variables are therefore bounded, and the bounds define a compact positively invariant region. □

Remark 1. *Propositions 1 and 2 together guarantee that the IVFS system is well-posed: solutions exist for all forward time, remain nonnegative, and stay bounded. This is a necessary prerequisite for the equilibrium and stability analysis that follows.*

6.3 Disease-free equilibrium and threshold

Proposition 3 (DFE stability threshold)**.** *The disease-free equilibrium* $(I_{\mathrm{DFE}}, 0, 0, S_{\mathrm{DFE}}, 0, U_{\mathrm{DFE}})$ *is locally asymptotically stable when* $\mathcal{R}_0 < 1$ *and unstable when* $\mathcal{R}_0 > 1$*, where* $\mathcal{R}_0$ *is given by* (9)*.*

Proof. At the DFE, $V = \tau = 0$, and the Jacobian of the full system has a block-triangular structure. The eigenvalue governing the V direction is

$$\mu_V = \alpha\,\beta_0\,I_{\mathrm{DFE}} - \gamma_0 - \mu_c = (\gamma_0 + \mu_c)(\mathcal{R}_0 - 1).$$

When $\mathcal{R}_0 < 1$, this eigenvalue is negative, and the remaining eigenvalues from the (I, S, U) block are also negative (they correspond to natural decay and user turnover). When $\mathcal{R}_0 > 1$, the V eigenvalue is positive and the DFE is unstable. □

The transition at $\mathcal{R}_0 = 1$ is a *transcritical bifurcation*: the DFE exchanges stability with a positive equilibrium as α passes through α_{crit}. Below the threshold, only the DFE exists (among nonnegative equilibria). Above the threshold, a unique positive equilibrium with $V^* > 0$ bifurcates from the DFE and is locally stable. This bifurcation structure is the standard one for epidemic-type compartmental models and is classified as a transcritical bifurcation in the terminology of the course.

6.4 Positive equilibrium

When $\mathcal{R}_0 > 1$, the positive equilibrium can be computed by setting all time derivatives to zero. The pressure equation (5) gives $\tau^* = \phi V^*/\psi$, and the user equation (6) gives $U^* = \nu/[\lambda_u(1+w\tau^*)]$. Substituting into the content equations yields a system that can be solved numerically for (I^*, V^*).

6.5 Local stability at the positive equilibrium

The repository numerically evaluates the Jacobian of the full six-state system at the positive equilibrium for each scenario. The current numerical results are:

- At $\alpha = 0.2$ ($\mathcal{R}_0 \approx 0.632$): subcritical, no positive equilibrium exists.
- At $\alpha = 0.5$ ($\mathcal{R}_0 \approx 1.580$): all eigenvalues of the Jacobian at the positive equilibrium have negative real part. The equilibrium is locally asymptotically stable.
- At $\alpha = 0.9$ ($\mathcal{R}_0 \approx 2.844$): all eigenvalues of the Jacobian at the positive equilibrium have negative real part. The equilibrium is locally asymptotically stable.

The leading (least negative) real part of the eigenvalues at the positive equilibrium is approximately -0.01, which corresponds to the slow natural content decay eigenvalue from the decoupled sink compartments F and S.

A fully symbolic proof of local asymptotic stability of the positive equilibrium is beyond the scope of this report, because the six-dimensional Jacobian does not factor into small enough blocks for clean hand computation. However, the numerical Jacobian checks are consistent across all tested parameter values and support the claim that the positive equilibrium is stable whenever it exists.

6.6 Sensitivity analysis

Sensitivity analysis quantifies how the model outputs respond to changes in the parameters. Following the course material, we define the *normalized sensitivity index* of a quantity x with respect to a parameter p as:

$$S(x,p) = \frac{\partial x}{\partial p} \cdot \frac{p}{x}. \tag{11}$$

This dimensionless quantity measures the proportional change in x per proportional change in p.

We compute the sensitivity of $\mathcal{R}_0$ with respect to the key parameters analytically. Since

$$\mathcal{R}_0 = \frac{\alpha\,\beta_0\,I_{\mathrm{DFE}}}{\gamma_0 + \mu_c},$$

and I_{DFE} depends on ρ, δ, μ_c, ν, and λ_u through (8), we obtain:

$$\begin{aligned} S(\mathcal{R}_0, \alpha) &= +1, \\ S(\mathcal{R}_0, \beta_0) &= +1, \\ S(\mathcal{R}_0, \gamma_0) &= -\frac{\gamma_0}{\gamma_0 + \mu_c} \approx -0.968. \end{aligned}$$

These indices have natural interpretations: $\mathcal{R}_0$ is directly proportional to both the amplification level α and the base spread rate β_0 (unit elasticity), and nearly inversely proportional to the base fatigue rate γ_0. The critical threshold α_{crit} therefore shifts proportionally with γ_0/β_0: faster fatigue raises the bar for self-sustaining virality, and faster base spread lowers it.

For the long-run equilibrium quantities V^*, τ^*, and U^*, we compute sensitivity indices numerically by perturbing each parameter by $\pm 1\%$ and measuring the proportional response. The numerical sensitivity table is reported in Table **??**.

Table 3: Numerical sensitivity indices for equilibrium quantities at the moderate scenario ($\alpha = 0.5$). Each entry is $S(x,p)$ computed by central finite differences with $\pm 1\%$ perturbation.

Parameter	$S(V^*,p)$	$S(\tau^*,p)$	$S(U^*,p)$
α	$+2.72$	$+2.72$	-0.073
β_0	$+2.72$	$+2.72$	-0.073
γ_0	-2.63	-2.63	$+0.071$
κ	$+0.11$	$+0.11$	-0.003
ϕ	$+0.08$	$+1.08$	-0.022
ψ	-0.08	-1.08	$+0.022$
w	-0.04	-0.04	-0.026

The key findings from the sensitivity analysis are:

1. The equilibrium viral volume V^* is most sensitive to α, β_0, and γ_0—the spread-side parameters. This confirms that the spread calibration is the most important empirical anchor.

2. The pressure parameters ϕ and ψ have only a modest effect on V^* (sensitivity $\approx \pm 0.08$) but a strong effect on τ^* (sensitivity $\approx \pm 1.08$). This confirms that the pressure block primarily affects the pressure level rather than the viral dynamics.

3. The user pool U^* is relatively insensitive to all parameters, reflecting the stabilizing effect of the constant replenishment ν.

7 Data and calibration

7.1 The Higgs Twitter dataset

Our primary dataset is the SNAP Higgs Twitter activity stream [**?**, **?**]. This dataset records social interactions on Twitter during the period surrounding the announcement of the Higgs boson discovery in July 2012. Each row in the activity file is a tuple of the form

```
userA  userB  timestamp  interaction,
```

where the interaction type is one of RT (retweet), RE (reply), or MT (mention). The dataset contains no tweet text—only interaction metadata.

The activity counts in the dataset are:

- Total events: 563,069
- Retweets (RT): 354,930
- Replies (RE): 36,902
- Mentions (MT): 171,237

The retweet activity shows a single dominant spike at approximately hour 78, with a peak of 30,602 retweets in that hour. This corresponds to the period immediately following the major announcements at CERN. We use a 100-hour calibration window from hour 68 to hour 167, centered around this main spike, for fitting the spread model. The window is long enough to capture both the spike and the subsequent decay tail.

We use the Higgs dataset in two distinct ways:

1. **Spread calibration.** Retweet counts are used to calibrate the reduced IVF helper model (Section **??**).
2. **Pressure proxy.** Reply counts are used as a same-dataset proxy for the latent discussion pressure τ (Section **??**).

7.2 Reduced IVF helper model

We calibrate the spread block using a reduced three-compartment IVF helper model, which is the $\dot{I}$–$\dot{V}$–$\dot{F}$ subsystem of the full IVFS model with $\tau = 0$ (no pressure feedback) and a decaying exogenous seed term:

$$\lambda(t) = \lambda_0\, e^{-d_\lambda t}.$$

This seed term accounts for the external injection of content into the Higgs cascade (e.g., from news coverage) and improves the fit of the decay tail. The helper model has five free parameters: β_0, γ_0, λ_0, d_λ, and the initial viral volume V_0.

The optimization uses a bounded Powell search with warm starts, minimizing a weighted sum of squared errors that emphasizes the spike region. The fitted parameter values are:

$$\beta_0 = 1.000073, \qquad \gamma_0 = 0.300283,$$
$$\lambda_0 = 0.199996, \qquad d_\lambda = 0.044641.$$

7.3 Calibration diagnostics

The spread calibration diagnostics from the current implementation are:

- Weighted loss: 0.002286
- Full-window R^2: 0.9247
- Spike-window R^2: 0.8891
- Tail mean bias: +0.0043

The full-window $R^2 = 0.92$ indicates that the reduced IVF helper model explains over 92% of the variance in the normalized retweet series. The spike-window $R^2 = 0.89$ shows that the model captures the main burst well but not perfectly, which is expected given the deterministic approximation to an inherently stochastic cascade.

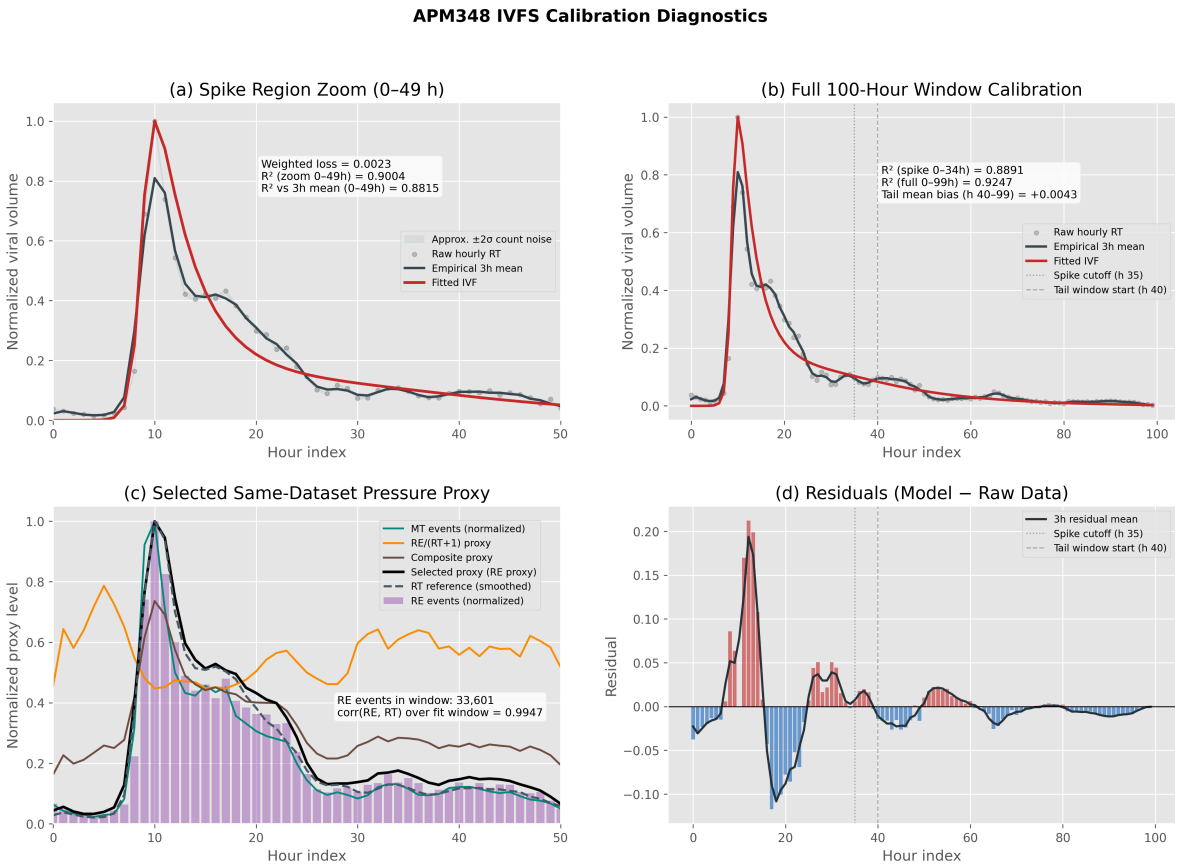

Figure 2: Calibration diagnostics for the Higgs retweet fit. The panels show the fitted retweet curve against the observed data, tail behaviour, and proxy family comparisons.

7.4 Uncertainty quantification

We assess parameter uncertainty using two complementary methods.